Sahaj Bhadja

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Summary

Backend Software Engineer with 2+ years building **Python** services where correctness matters: **idempotent** payment webhooks, **reconciliation** pipelines, and failure-isolated jobs, plus **blockchain** research on consensus and Web3 wallet integrations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built SQL transformation and **reconciliation** pipelines across PostgreSQL, SQL Server, and cloud storage with incremental loads and validation checks, running 5K+ monthly jobs at 99%+ production reliability.
- Designed explicit job-state transitions and **failure isolation** for async FastAPI workers, processing 1K+ AI jobs at 3x throughput, and normalized multi-tenant schemas that cut dashboard latency from 1.2s to 350ms.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built Python Flask **backend** services and REST APIs for 150+ instructors with normalized MySQL persistence on AWS RDS, handling 10K+ daily requests with stable latency and 60% faster feedback.
- Implemented concurrent batch processing with fault-tolerant execution and timeout-aware model fallback, logging every call's tokens and cost to an **auditable** usage table across 1K+ files per batch.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Optimized large-scale API integrations with indexing, caching, pagination, and field filtering, improving response times 30% and cutting technical debt 20% through object-oriented **Singleton and Factory** module design.
- Wrote 20+ **unit and integration test** suites with PyTest and Mockito in nightly Jenkins CI/CD and led an AWS Glue and S3 ETL redesign that boosted pipeline efficiency 68%.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Designed a **quantum-resistant blockchain** pairing lattice-based FALCON signatures with node-monitored delegated proof-of-stake consensus and Borda Count witness selection, cutting centralized-control and quantum-risk exposure 25% in benchmark simulations.
- **Benchmarked** the protocol against an ECDSA baseline across 100–1,000 transaction workloads, sustaining 957.8 TPS at 224ms versus 968.3 TPS and 211ms, and presented results at IISc Bengaluru.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Integrated PCI-compliant multi-currency payments with fraud checks and **idempotent** webhooks for refunds, disputes, and chargebacks, keeping payment status consistent across retries and cutting manual payment handling by 70%.
- Implemented **wallet-signature** authentication with web3.js and MetaMask, eliminating password storage for wallet users, and used Firebase transactional locking to prevent double-booked time slots across 50K+ booking records.

Skills

Languages & Backend: Python, Java, TypeScript, JavaScript, SQL, Solidity, FastAPI, Flask, Node.js, REST APIs
Correctness: Idempotency, Reconciliation, Data Validation, Data Modeling, Transactional Locking, Webhooks, Async Job Orchestration
Data & Cloud: PostgreSQL, SQL Server, MongoDB, Firebase, AWS, Azure, Docker, Kubernetes, CI/CD
Blockchain: Smart Contracts, web3.js, MetaMask, Truffle, Ganache, Consensus Algorithms, Lattice-Based Cryptography
Practices: System Design, Microservices, Jenkins, GitHub Actions, PyTest, Unit and Integration Testing, Wallet-Signature Auth

Projects

On-Chain Transfer Indexer and Reconciler | *Python, web3.py, Solidity, PostgreSQL, Docker*

- Designed a Python **EVM indexer** that ingests ERC-20 transfer logs over JSON-RPC with idempotent upserts keyed by transaction hash and log index, handles reorgs via confirmation depth, and **reconciles** balances in Postgres.

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid, Kubernetes*

- Built a fintech platform in TypeScript and Next.js with Plaid bank linking, a token-based firewall over **transaction** access, and real-time updates across 1M+ transactions, deployed on Kubernetes with CI/CD.

Comprehensive Analysis of Blockchain for User Authentication Paper | *Smart Contracts, DLT, Self-Sovereign Identity*

- Co-authored an IEEE paper analyzing **smart contract** authentication on distributed ledgers, including self-sovereign identity, IoT device identity, role-based access, and replay, man-in-the-middle, and phishing threats to on-chain verification.